

# Mixed Factoring: Choosing and Combining Methods

## Answer Key

### Algebra 1 Polynomials

#### Quick Answers

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|----------------------|---------------------------------|
| 1. $3x(2x - 5)$      | 7. $(2x - 3)^2$                 |
| 2. $(x + 4)(x + 5)$  | 8. $2x(x - 3)(x + 3)$           |
| 3. $(x - 8)(x + 8)$  | 9. $(2x - 3)(3x + 1)$           |
| 4. $(x + 3)(3x - 9)$ | 10. $2x^2 + 8x + 8, 2(x + 2)^2$ |
| 5. $(x + 3)(2x + 1)$ | 11. $(x - 3)(x + 3)(x^2 + 9)$   |
| 6. $(x - 7)(x + 4)$  | 12. $4x(x - 2)(x + 3)$          |
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#### Curriculum

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**Level:** Algebra 1 Polynomials

HSA-APR.A.1 – *problems 8, 10*

HSA-SSE.B.3 – *problems 1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 12*

*Difficulty 1–5 of 5 across 14 tagged checks.*

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#### 1. Factor completely: $6x^2 - 15x$ .

**Step 1.** Checklist step one: is there a common factor? The coefficients 6 and 15 share 3, and both terms carry an  $x$ , so the greatest common factor is  $3x$ .

**Step 2.**  $6x^2 - 15x = 3x(2x - 5)$ .

**Step 3.** The bracket has two terms but no subtraction of two squares, and it is degree one, so nothing factors further. Technique: GCF only.

$3x(2x - 5)$

#### 2. Factor completely: $x^2 + 9x + 20$ .

**Step 1.** No common factor. Three terms with leading coefficient 1, so look for a factor pair of 20 that adds to 9.

**Step 2.** 4 and 5:  $4 \cdot 5 = 20$  and  $4 + 5 = 9$  ✓

**Step 3.**  $x^2 + 9x + 20 = (x + 4)(x + 5)$ . Technique: monic trinomial.

$(x + 4)(x + 5)$

#### 3. Factor completely: $x^2 - 64$ .

**Step 1.** No common factor. Two terms with a subtraction sign between them: test the difference of squares.

**Step 2.**  $x^2 = (x)^2$  and  $64 = (8)^2$ , so  $A = x$  and  $B = 8$ .

**Step 3.**  $A^2 - B^2 = (A - B)(A + B)$  gives  $(x - 8)(x + 8)$ . Expanding back, the middle terms  $+8x$  and  $-8x$  cancel ✓ Technique: difference of squares.

$(x - 8)(x + 8)$

**4. Factor completely:**  $3x^2 - 27$ .

**Step 1.** It looks like a difference of squares, but  $3x^2$  is not a perfect square, so the pattern does not apply yet. Checklist step one first: both terms are divisible by 3.

**Step 2.**  $3x^2 - 27 = 3(x^2 - 9)$ .

**Step 3.** Now the bracket *is* a difference of squares with  $A = x$  and  $B = 3$ , so  $x^2 - 9 = (x - 3)(x + 3)$  and the complete factorization is  $3(x - 3)(x + 3)$ . Technique: GCF, then difference of squares.

$$3(x - 3)(x + 3)$$

**5. Factor completely:**  $2x^2 + 7x + 3$ .

**Step 1.** No common factor (3 is odd), and the leading coefficient is 2, not 1 — so factor by grouping on the product of the outer coefficients.

**Step 2.** That product is  $2 \cdot 3 = 6$ , and we need a factor pair of 6 adding to 7: those are 6 and 1. Split the middle term:  $2x^2 + 6x + x + 3$ .

**Step 3.** Group:  $2x(x + 3) + 1(x + 3) = (2x + 1)(x + 3)$ . Technique: non-monic trinomial by grouping.

$$(2x + 1)(x + 3)$$

**6. Flower bed of area  $x^2 - 3x - 28$  square feet.**

**Step 1.** A rectangle's area is length times width, so factoring the area into two brackets is exactly what produces the two side lengths. No common factor, leading coefficient 1: look for a factor pair of  $-28$  that adds to  $-3$ .

**Step 2.** The constant is negative, so the pair has opposite signs and the larger one is negative:  $-7$  and  $+4$  give  $-7 \cdot 4 = -28$  and  $-7 + 4 = -3$  ✓

**Step 3.**  $x^2 - 3x - 28 = (x - 7)(x + 4)$ , so the bed measures  $x - 7$  feet by  $x + 4$  feet. Technique: monic trinomial.

$$(x - 7)(x + 4)$$

**7. Factor completely:**  $4x^2 - 12x + 9$ .

**Step 1.** No common factor (9 is odd). Three terms whose first and last are perfect squares — test the perfect-square pattern before trying pairs.

**Step 2.**  $4x^2 = (2x)^2$  and  $9 = (3)^2$ , so  $A = 2x$  and  $B = 3$ . The pattern demands a middle term of  $2AB = 2 \cdot 2x \cdot 3 = 12x$ , and the trinomial has  $-12x$ : the size matches and the sign is negative.

**Step 3.** A negative middle term means the bracket is a difference:  $4x^2 - 12x + 9 = (2x - 3)^2$ . Technique: perfect-square trinomial.

$$(2x - 3)^2$$

**8. Factor completely:**  $2x^3 - 18x$ .

**Step 1.** Two terms, but neither is a perfect square as written — and both are divisible by 2 and both contain  $x$ . The greatest common factor is  $2x$ .

**Step 2.**  $2x^3 - 18x = 2x(x^2 - 9)$ .

**Step 3.** The bracket is a difference of squares with  $A = x$ ,  $B = 3$ :  $x^2 - 9 = (x - 3)(x + 3)$ . Complete:  $2x(x - 3)(x + 3)$ . Technique: GCF, then difference of squares.

$$2x(x - 3)(x + 3)$$

**9. Factor completely:**  $6x^2 - 7x - 3$ .

**Step 1.** No common factor, leading coefficient 6: grouping again. The product of the outer coefficients is  $6 \cdot (-3) = -18$ .

**Step 2.** A factor pair of  $-18$  adding to  $-7$  is  $-9$  and  $+2$ . Split the middle term:  $6x^2 - 9x + 2x - 3$ .

**Step 3.** Group:  $3x(2x - 3) + 1(2x - 3) = (2x - 3)(3x + 1)$ . Expanding back gives  $6x^2 + 2x - 9x - 3 = 6x^2 - 7x - 3$  ✓ Technique: non-monic trinomial by grouping.  $\boxed{(2x - 3)(3x + 1)}$

**10. Error analysis: is  $2x^2 + 8x + 8 = (2x + 4)(x + 2)$  complete?**

**Step 1 (part a).** Multiply the student's answer out:  $(2x + 4)(x + 2) = 2x^2 + 4x + 4x + 8 = 2x^2 + 8x + 8$ . It really does equal the original, so expanding alone cannot expose the fault — this is why “completely” matters.

**Step 2 (part b).** The first bracket still has a common factor:  $2x + 4 = 2(x + 2)$ . The student skipped checklist step one, so a factor of 2 was left buried inside a bracket instead of pulled to the front.

**Step 3 (part c).** Take the GCF out first:  $2x^2 + 8x + 8 = 2(x^2 + 4x + 4)$ , and  $x^2 + 4x + 4$  is a perfect square,  $(x + 2)^2$ . So the complete factorization is  $2(x + 2)^2$ .

*Manual review (part b):* accept any sentence saying that  $2x + 4$  still contains a common factor of 2, or equivalently that the answer is a correct product but not fully factored.

$$\boxed{2x^2 + 8x + 8 \quad \text{and} \quad 2(x + 2)^2}$$

**11. Factor completely:**  $x^4 - 81$ .

**Step 1.** No common factor. Two terms with a subtraction sign, and both are squares:  $x^4 = (x^2)^2$  and  $81 = (9)^2$ .

**Step 2.** Difference of squares with  $A = x^2$ ,  $B = 9$ :  $x^4 - 81 = (x^2 - 9)(x^2 + 9)$ .

**Step 3.** Check each bracket again.  $x^2 - 9$  is itself a difference of squares,  $(x - 3)(x + 3)$ ;  $x^2 + 9$  is a *sum* of squares and does not factor over the integers, so the work stops there.

Technique: difference of squares, twice.

$$\boxed{(x - 3)(x + 3)(x^2 + 9)}$$

**12. Synthesis challenge: factor completely**  $4x^3 + 4x^2 - 24x$ .

**Step 1.** Checklist step one: the coefficients 4, 4 and 24 all share 4, and every term carries an  $x$ , so the greatest common factor is  $4x$ .

**Step 2.**  $4x^3 + 4x^2 - 24x = 4x(x^2 + x - 6)$ . The bracket now has leading coefficient 1, which is the easy trinomial case — taking the GCF out first is what turned a messy cubic into it.

**Step 3.** A factor pair of  $-6$  adding to  $+1$  is  $+3$  and  $-2$ , so  $x^2 + x - 6 = (x + 3)(x - 2)$  and the complete factorization is  $4x(x + 3)(x - 2)$ . Technique: GCF, then monic trinomial.

$$\boxed{4x(x + 3)(x - 2)}$$